
Group Quantum Key Distribution for HAP-Aided IoT Networks over FSO Channels

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1. Introduction

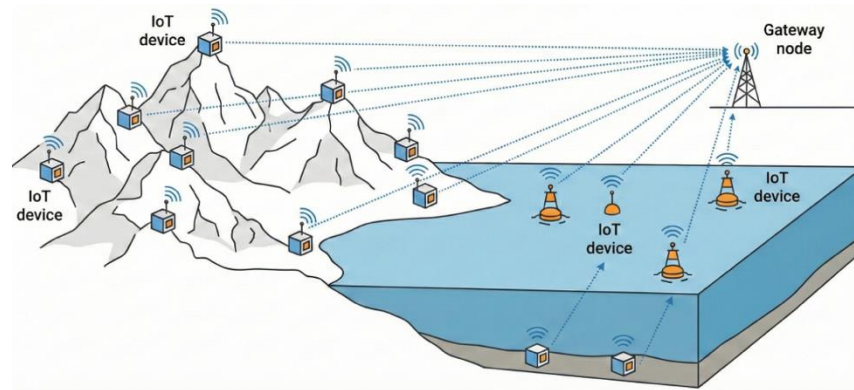
Background: Security of IoT Networks

Rapid IoT Growth : Connecting remote/resource-constrained devices

- Simultaneous IoT Device Updates
- Real-Time Device-to-Device Coordination

The Threat of Quantum Computers: Classical cryptography is vulnerable to future quantum computer attacks.

Scalable and efficient distribution of a quantum-resistant **group key** to securely share data among numerous devices in wide-area networks.



Research Motivation: Conventional Group Key Protocol

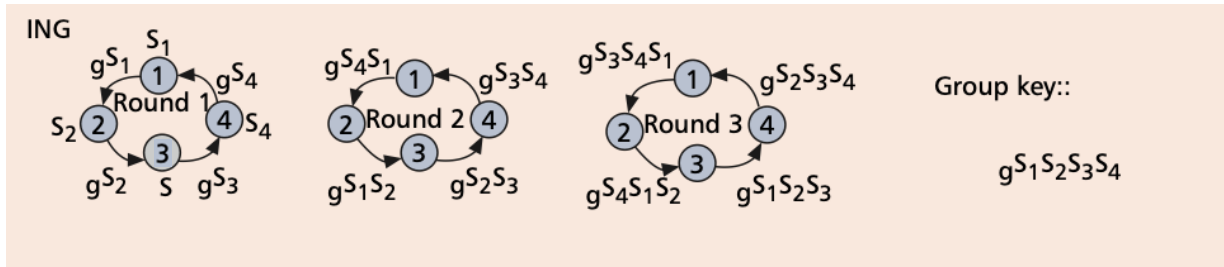
Conventional Group Key (e.g., ING)

Mechanism: Nodes form a "Logical Ring" to share partial keys sequentially.

Scalability: Establishing a group key requires **M-1 rounds**.

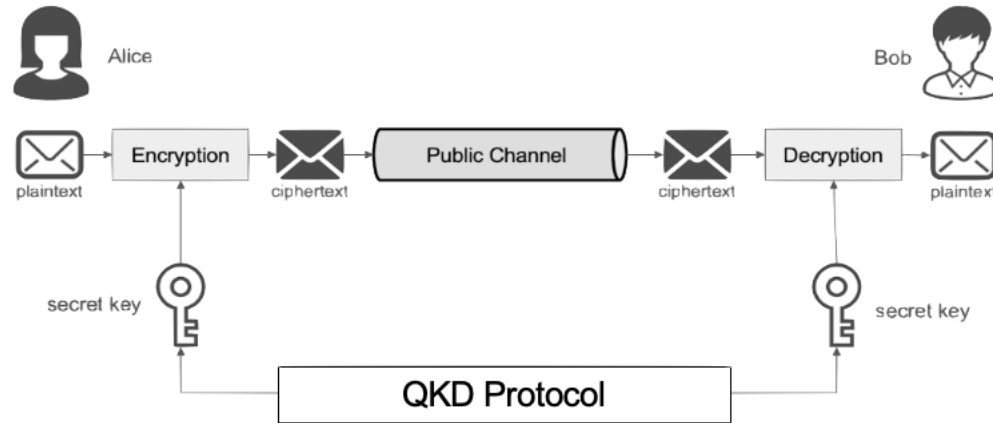
High Latency: For massive IoT networks, this step-by-step process causes long delays.

The Threat of Quantum Computers: Relies on traditional mathematical complexity, which will be easily broken by future quantum computers.



Quantum Key Distribution

QKD is a secure communication technology to distribute secret keys.



Extension to Group Key Distribution: NGHZ-QKD

- A specialized protocol for direct group key generation using multi-partite entanglement.

Research Motivation - Limitations

Current QKD Implementation

- Relies on **Optical Fiber** channels.
- Suitable for static networks between central servers (e.g., Banks).

Challenges in IoT Environment

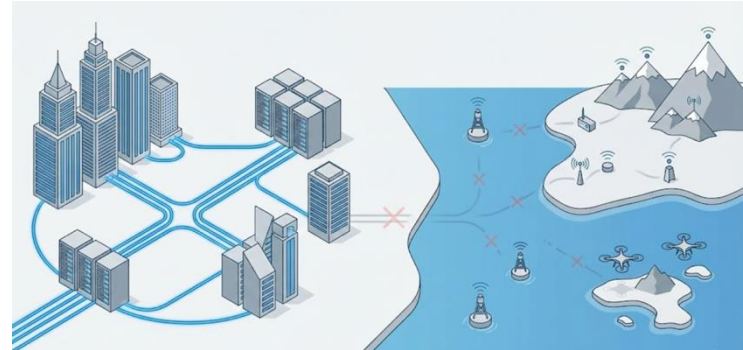
- IoT devices are distributed in **remote areas** (e.g., mountains, islands).
- Laying optical fiber to every device is **costly and impractical**.

→ **Fiber-based QKD may not be feasible for the IoT networks.**

Current NGHZ-QKD Limitations

- Generating multiphoton quantum entanglement states (e.g., 10 GHZ) is difficult.
- Currently in the experimental stage.

→ **Not currently feasible.**



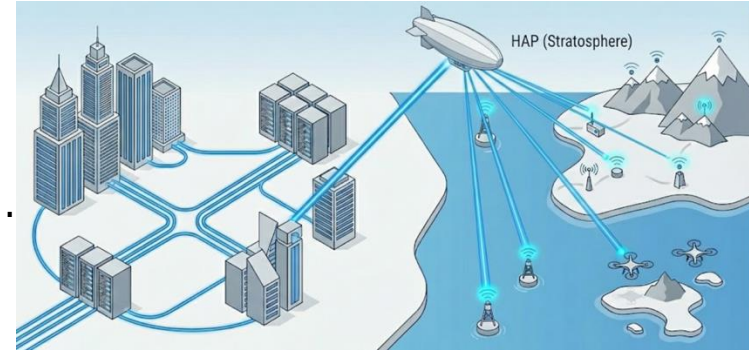
Research Motivation - Proposed Solution

Wireless Solution: Free-Space Optics (FSO)

- Replaces physical fiber with **wireless FSO links**.

High-Altitude Platform (HAP)

- Operates in the **stratosphere** (Approximately 20km).
- Acts to cover **wide areas** and **remote regions**.
- Supports ground devices without physical fiber.



Small-scale GHZ-QKD

- Generating small-scale GHZ is practically achievable.
- Generate keys per small group via GHZ-QKD, then use them to encrypt and distribute the group key.
- By processing several nodes simultaneously, a reduction in the overall processing steps can be expected.

Research Goals

- This work proposes an HAP-based FSO GHZ-QKD system to distribute the group key for multiple IoT nodes.
 1. To evaluate the design's performance, we develop a Qiskit-based simulation framework that incorporates major adverse effects in FSO channels.
 2. Comparing the performance of NGHZ-QKD with that of 2QKD
 3. Exploring the advantages of NGHZ-QKD in group key distribution.

2. System Description

System Model

QKD Protocol:

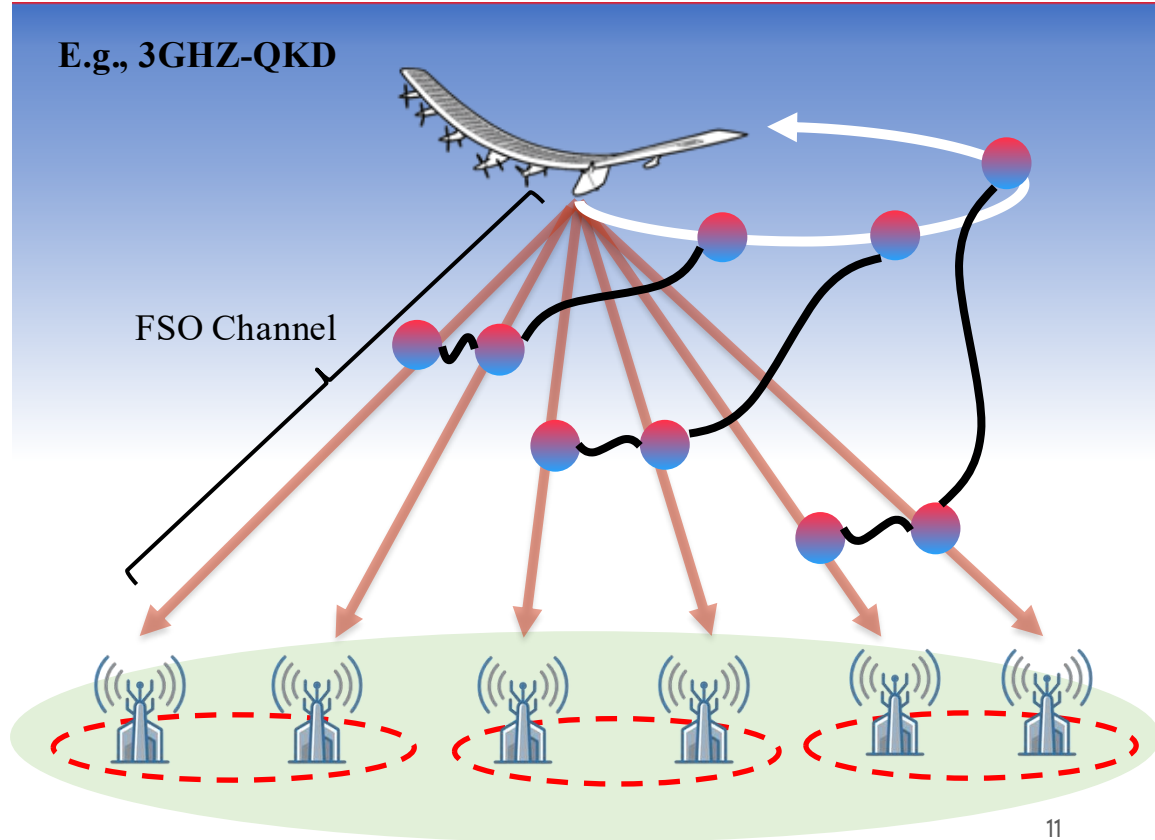
GHZ-QKD for generating the group keys.

HAP:

Generate photons in the GHZ state, retains one itself, and distributes the rest to the ground.

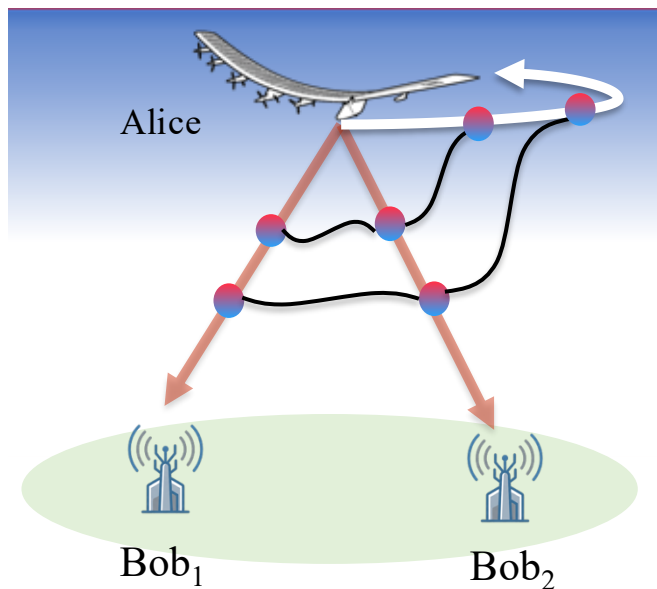
Ground Nodes:

The group is separated from other ground nodes, and photons are received by each group using TDMA.



3GHZ-QKD

$$|\phi_3\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)$$



	Photon				
Alice	Basis				
	Bit	1	0	1	0

	Photon				
Bob ₁	Basis				
	Bit	1	0	1	1

	Photon				
Bob ₂	Basis				
	Bit	1	0	1	1

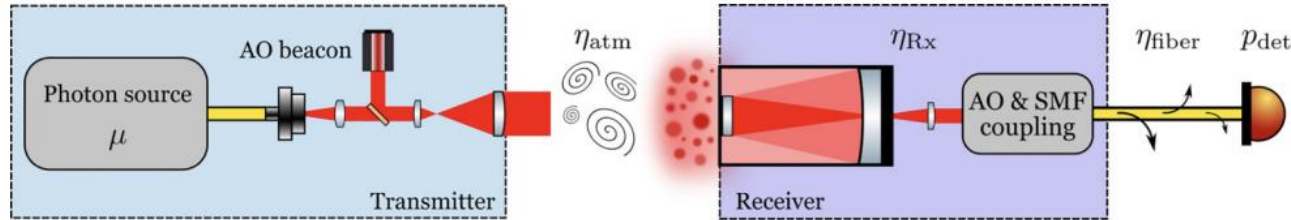
Record "11" as the sifted key

Adverse issues in FSO Channels

Photons are transmitted through free-space optics (FSO) channel

$$\eta_{free_space} = \eta_{atm} \times \eta_{R_X}$$

$$\eta_{R_X} = \eta_{DR_X} \times \eta_{SMF}$$



η_{atm} : Atmospheric Transmittance

η_{R_X} : The losses at the receiver

η_{DR_X} : The receiver collection efficiency

η_{SMF} : The fiber-coupling efficiency

These factors cause photon loss and photon state changes

→ **Affect key generation**

Group Key Distribution and Equation using TDMA

$$GroupSKR = \frac{\min(SKR_1 \cdot \Delta t, SKR_2 \cdot \Delta t, \dots, SKR_G \cdot \Delta t)}{G \cdot (\Delta t + d_{pp} + t_{enc} + d_{prop})}$$

$$SKR_i = R_{rep} \cdot Y_N \cdot r_{\infty} \cdot P_{sift}$$

$$G = \left\lceil \frac{M}{N-1} \right\rceil$$

Δt : Quantum phase duration

d_{pp} : Post-processing delay

t_{enc} : Encryption time

d_{prop} : Propagation delay

G : # of Group

M : # of Users

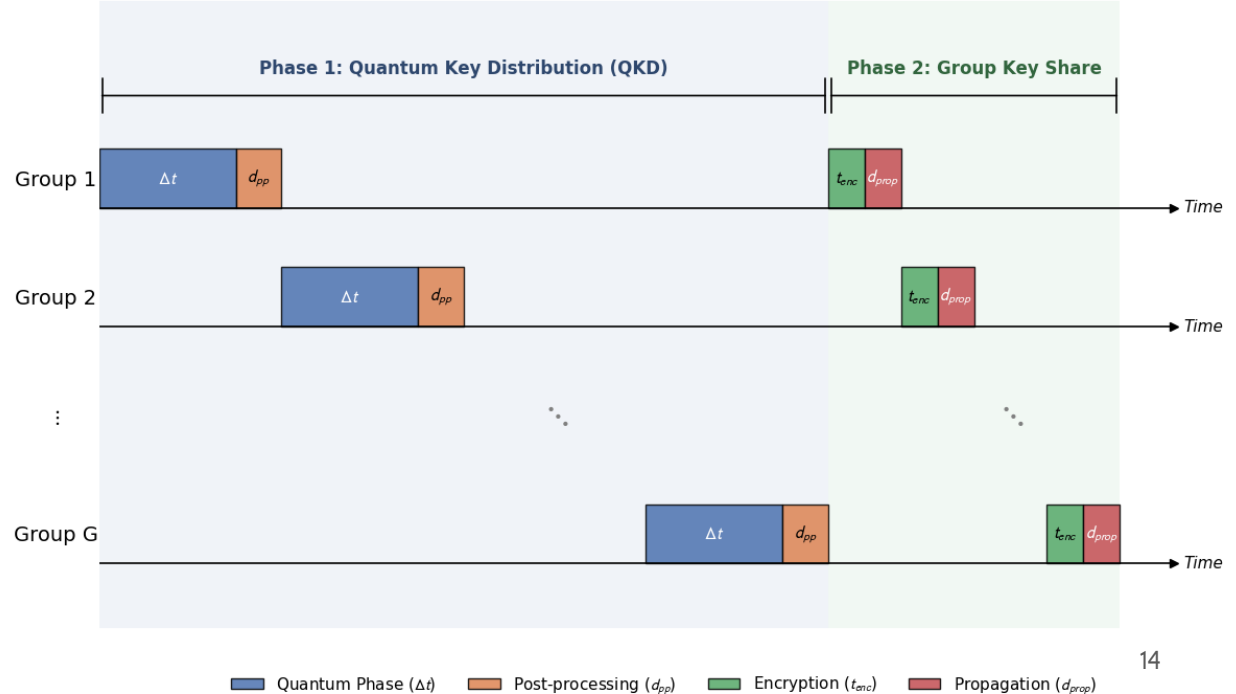
N : Degree of GHZ state

R_{rep} : Pulse repetition rate

Y_N : Probability of detection event

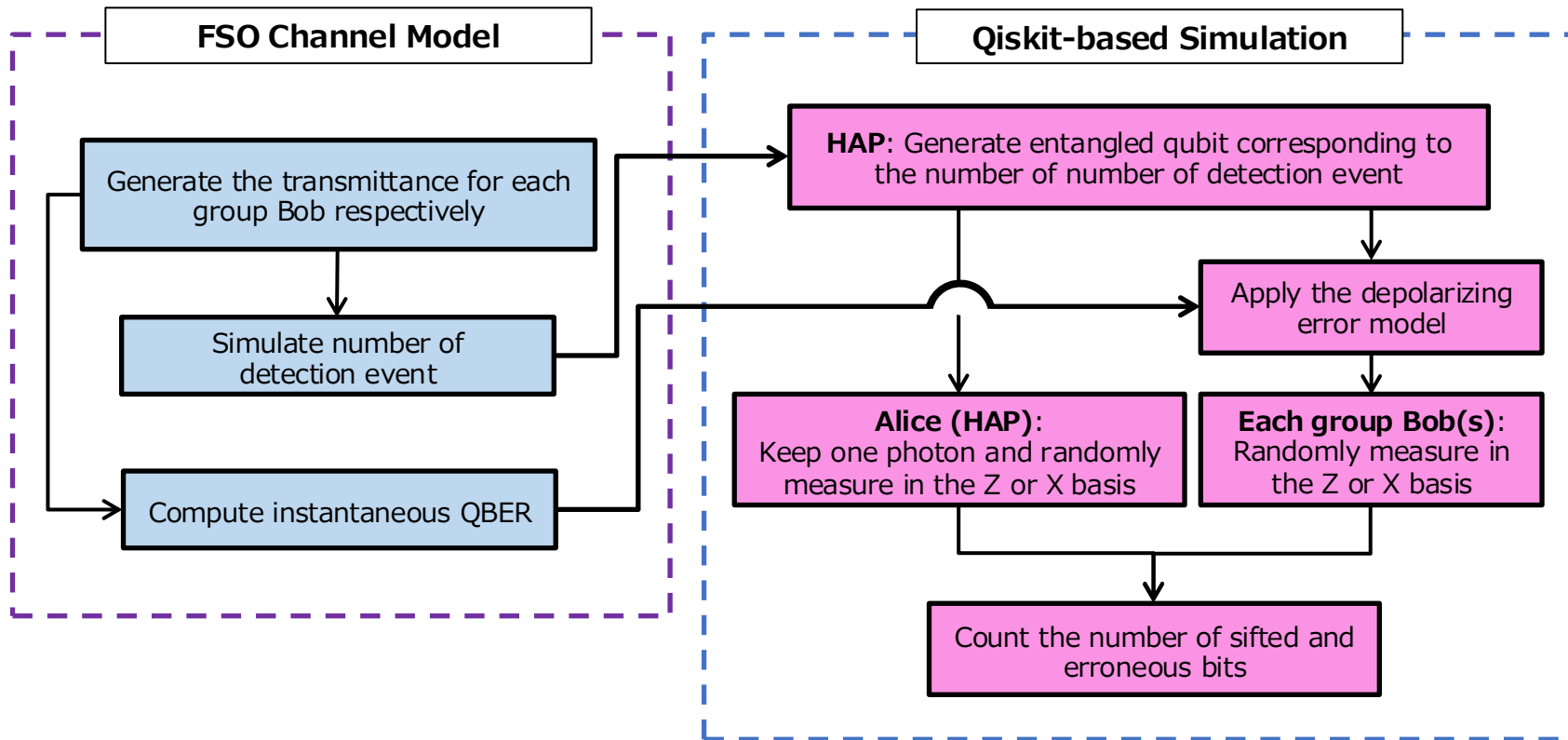
r_{∞} : Secret bit fraction & Shared states

P_{sift} : Sifting probability



Qiskit-based Simulation Framework

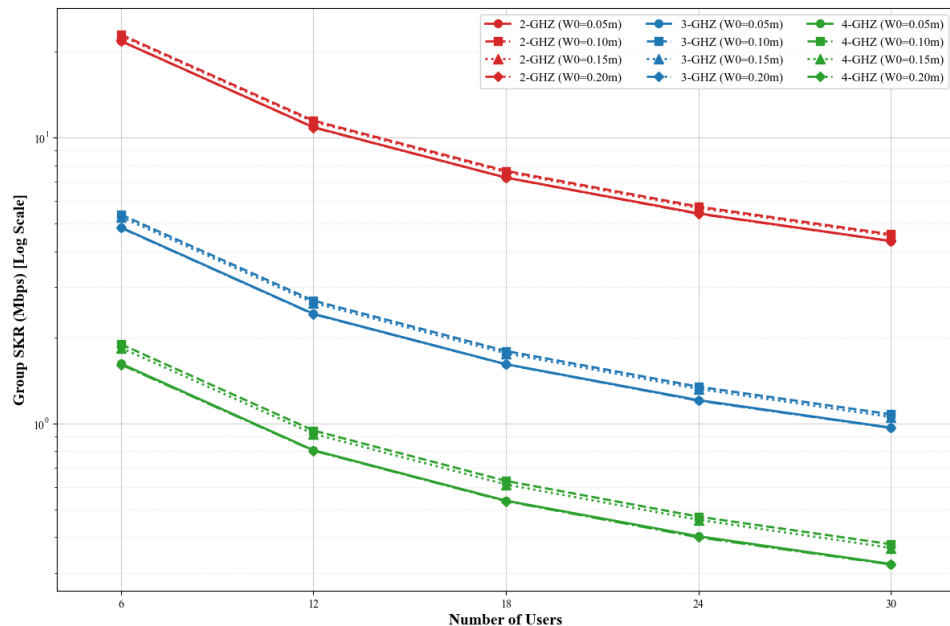
Qiskit: An open-source SDK provided by IBM for quantum computing.



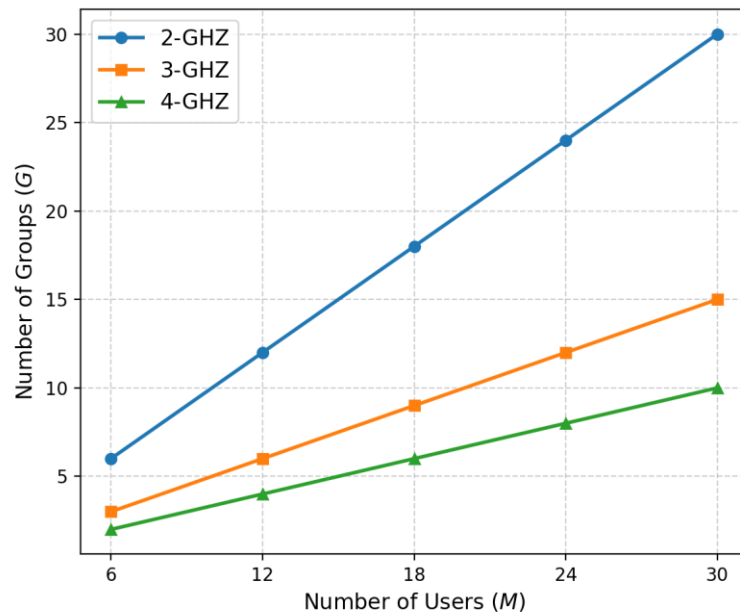
3. Results

Result 1: Group SKR and Number of Groups Comparison

Group SKR by different beam waist



Number of groups per GHz state



Result 2: Adjusting Shifting Probability

2GHZ (BBM92) shifting probability:

- Matching Z basis or X basis
- The probability of coincidence is $\frac{1}{2}$

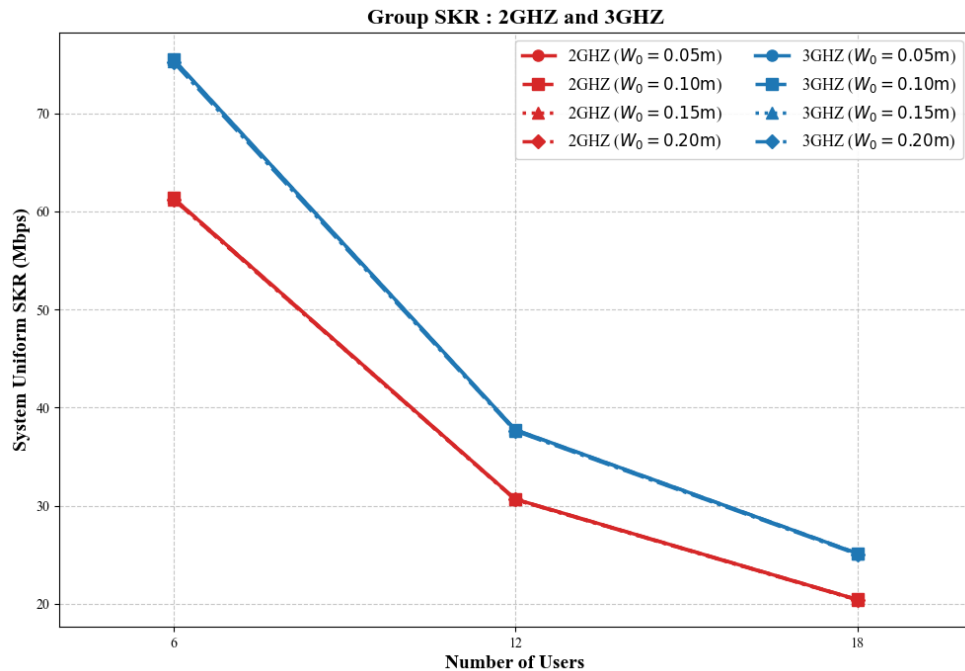
NGHZ shifting probability:

- Matching only Z basis
- The probability of coincidence is $\left(\frac{1}{2}\right)^N$

→ Difficult for GHZ-QKD to achieve a high SKR.

Proposal: Adjusting shifting probability

- Set the Z basis probability to 90%
- Existing studies on adjusted shifting probabilities.





NQKD(GHZ)

$$|GHZ\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)$$

